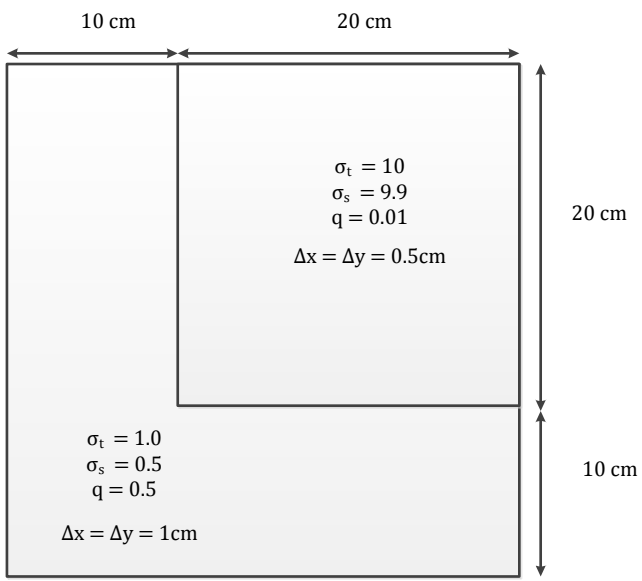
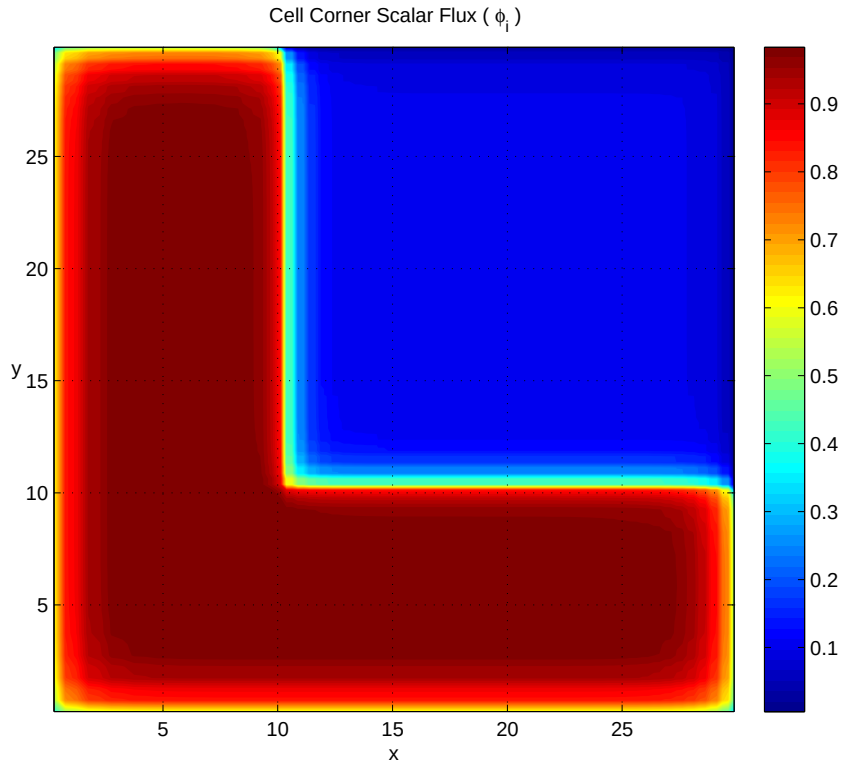
Definition of Test B



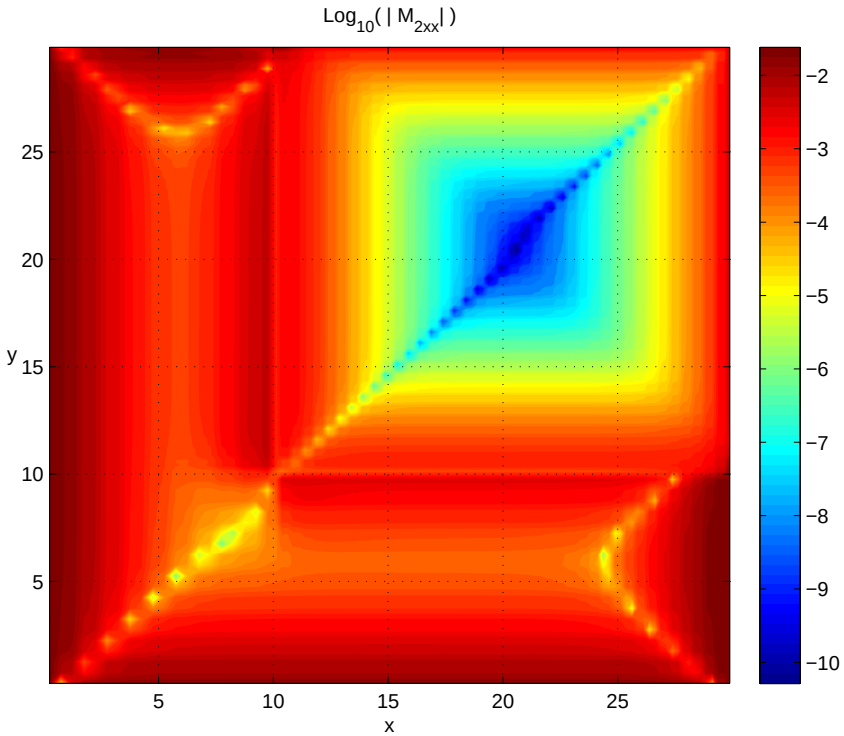
Map



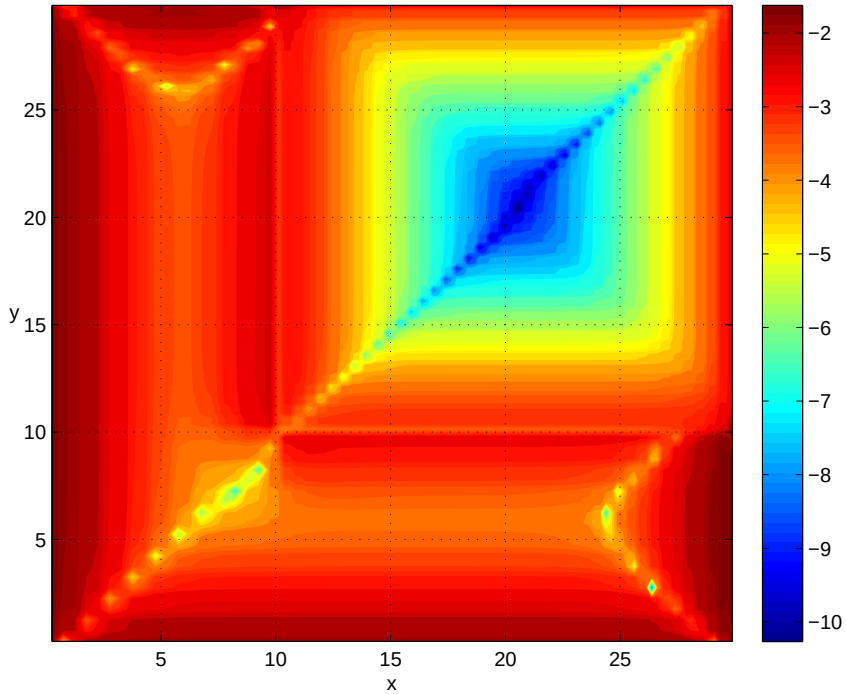
Solution

- spatial mesh:
  - $0 \leq x, y \leq 10$ cm: $\Delta x = \Delta y = 1$ cm, $\sigma_t \Delta x = \sigma_t \Delta y = 1$
  - $10 \leq x, y \leq 30$ cm: $\Delta x = \Delta y = 0.5$ cm, $\sigma_t \Delta x = \sigma_t \Delta y = 5$
- Abu-Shumays' quadruple-range quadratures with 36 angles/octant

Metric $M^E_{xx} = \left|\frac{1}{3} - E_{xx}\right|$



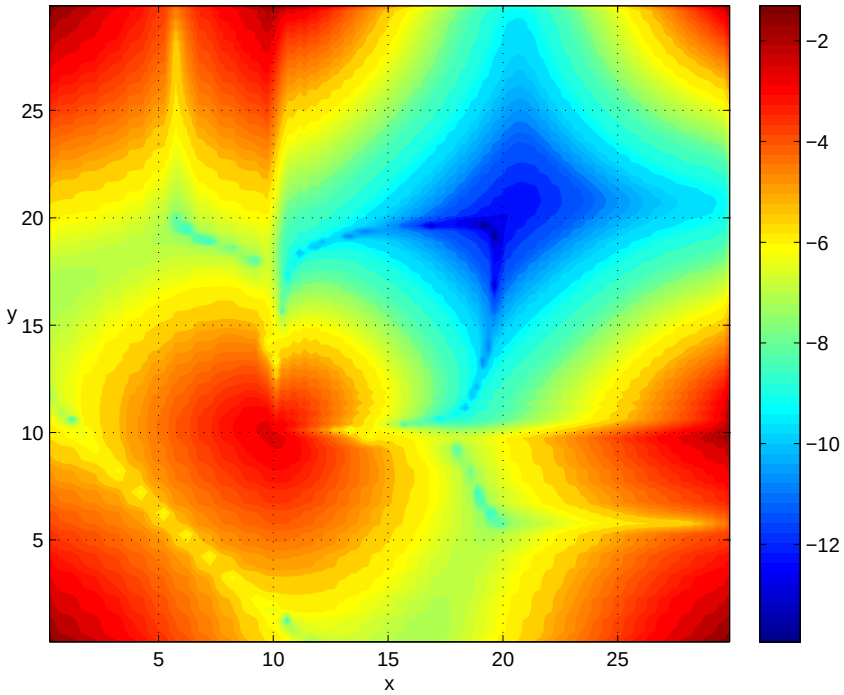
Converged

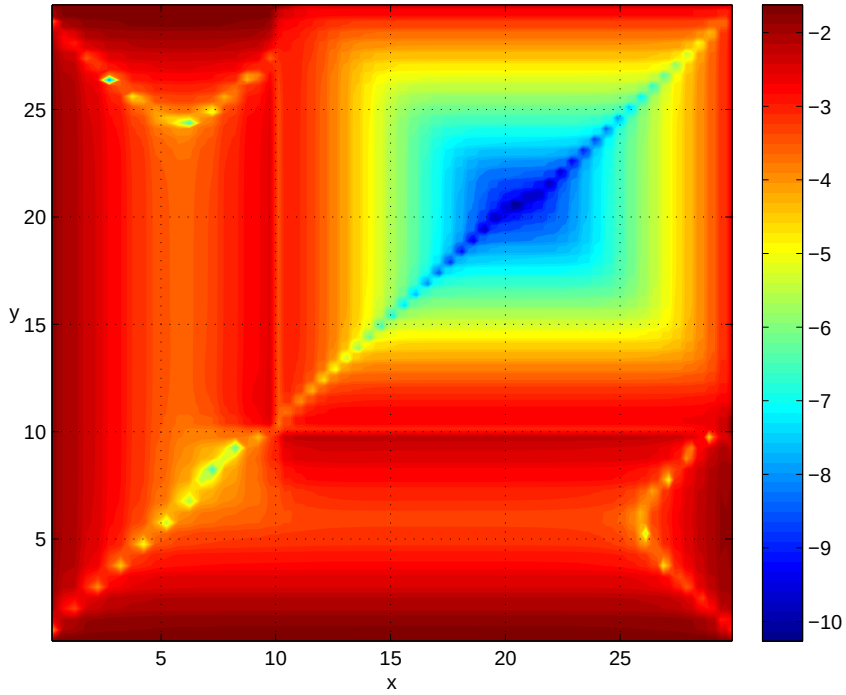


Estimated

- Log scale

Metric M^E (Estimated)



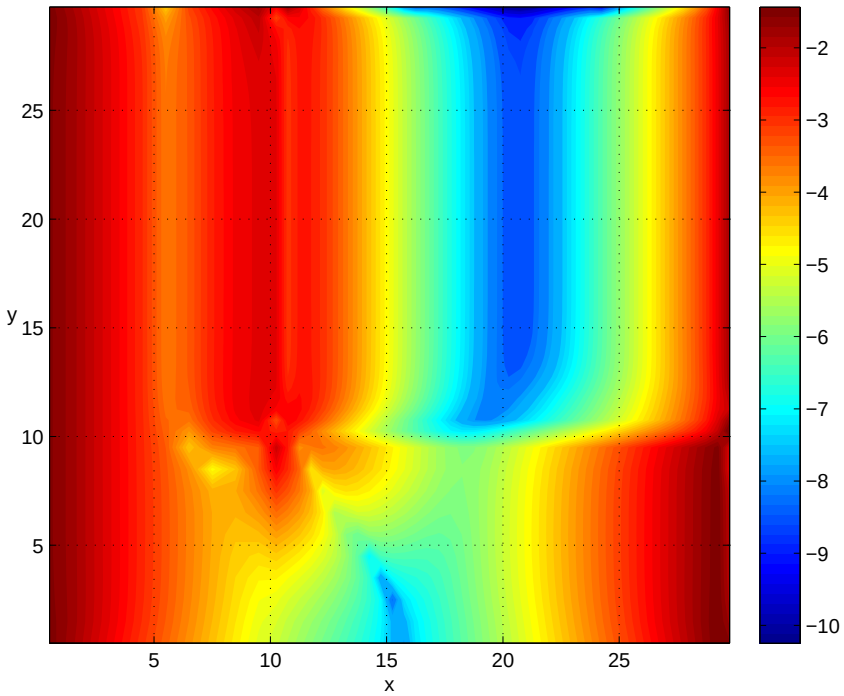


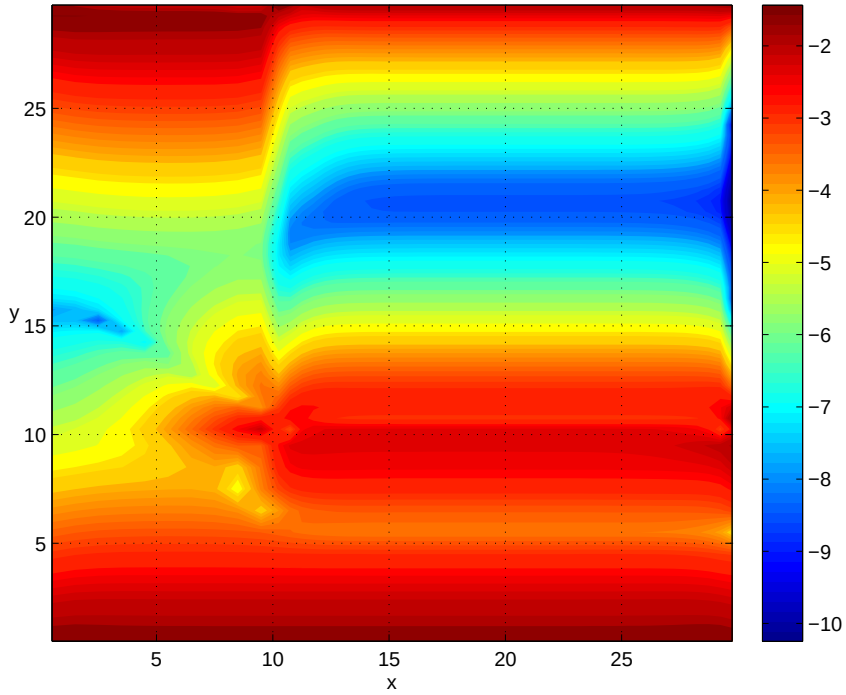
$$\mathbf{M}^{\mathbf{E}}_{\mathbf{xy}} = |\mathbf{E}_{\mathbf{xy}}|$$

$$\mathbf{M}^{\mathbf{E}}_{\mathbf{yy}} = \left|\frac{\mathbf{1}}{\mathbf{3}} - \mathbf{E}_{\mathbf{yy}}\right|$$

- Log scale

Metric $M^{\partial E}$ (Estimated)



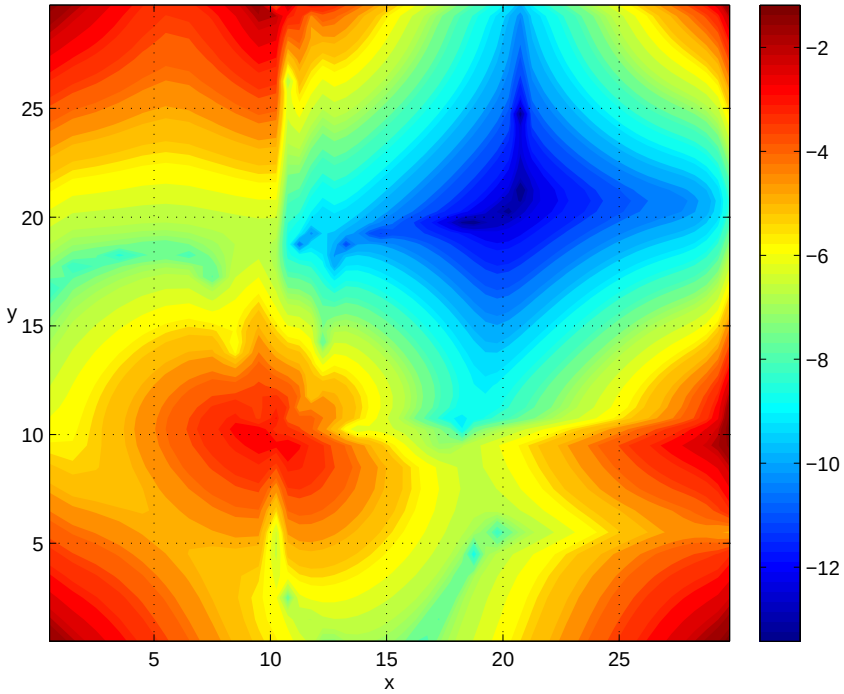


$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{xx}} = \left| \frac{\partial \mathbf{E}_{\mathbf{xx}}}{\partial \mathbf{x}} \right|$$

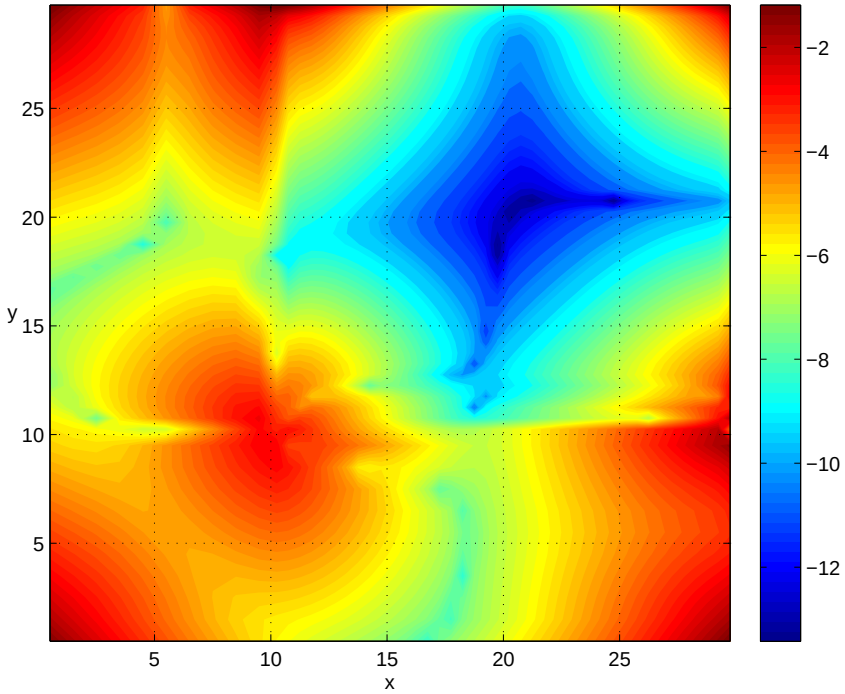
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{yy}} = \left| \frac{\partial \mathbf{E}_{\mathbf{yy}}}{\partial \mathbf{y}} \right|$$

- Log scale

Metric $M^{\partial E}$ (Estimated)



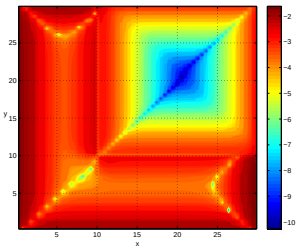
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{xy}} = \left| \frac{\partial \mathbf{E}_{\mathbf{xy}}}{\partial \mathbf{x}} \right|$$



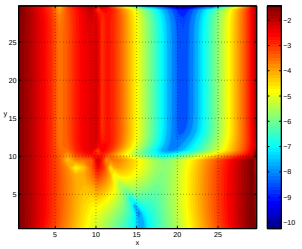
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{yx}} = \left| \frac{\partial \mathbf{E}_{\mathbf{xy}}}{\partial \mathbf{y}} \right|$$

- Log scale

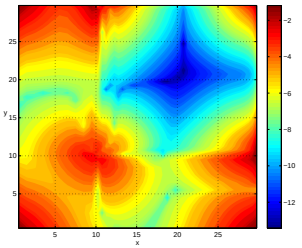
A Set of Metrics



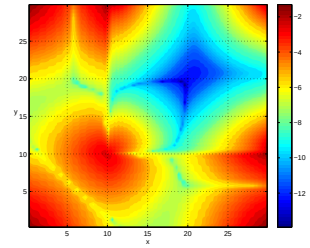
$$\mathbf{M}^{\mathbf{E}}_{\mathbf{xx}} = \left|\frac{\mathbf{1}}{\mathbf{3}} - \mathbf{E}_{\mathbf{xx}}\right|$$



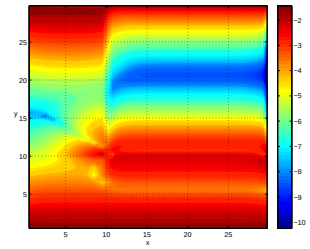
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{xx}} = \left|\frac{\partial \mathbf{E}_{\mathbf{xx}}}{\partial \mathbf{x}}\right|$$



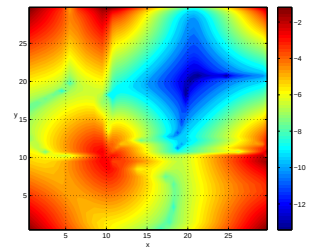
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{xy}} = \left|\frac{\partial \mathbf{E}_{\mathbf{xy}}}{\partial \mathbf{x}}\right|$$



$$\mathbf{M}^{\mathbf{E}}_{\mathbf{xy}} = \left|\mathbf{E}_{\mathbf{xy}}\right|$$



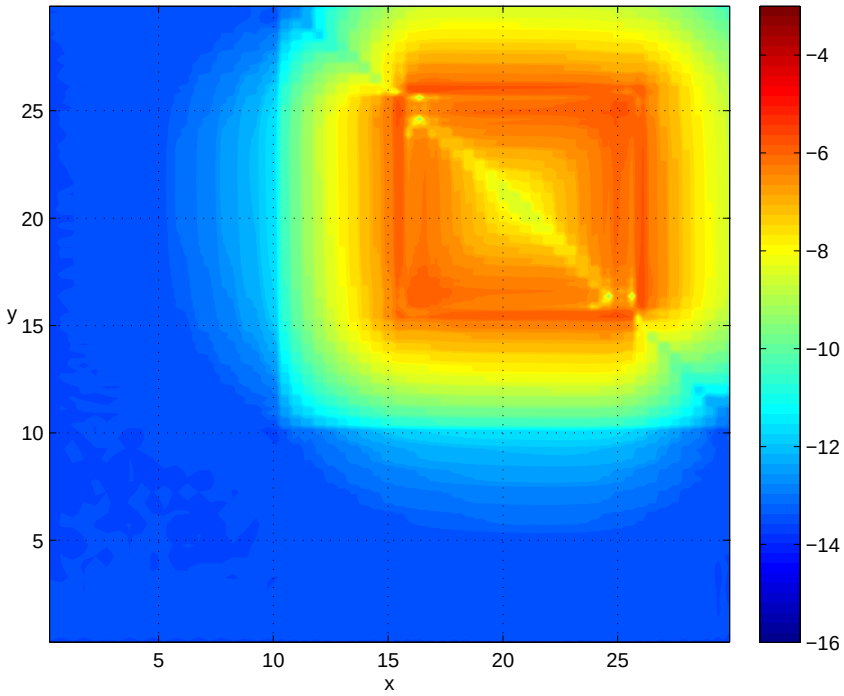
$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{yy}} = \left|\frac{\partial \mathbf{E}_{\mathbf{yy}}}{\partial \mathbf{y}}\right|$$



$$\mathbf{M}^{\partial \mathbf{E}}_{\mathbf{yx}} = \left|\frac{\partial \mathbf{E}_{\mathbf{xy}}}{\partial \mathbf{y}}\right|$$

Relative Error in Reproducing the Solution on a Given Grid

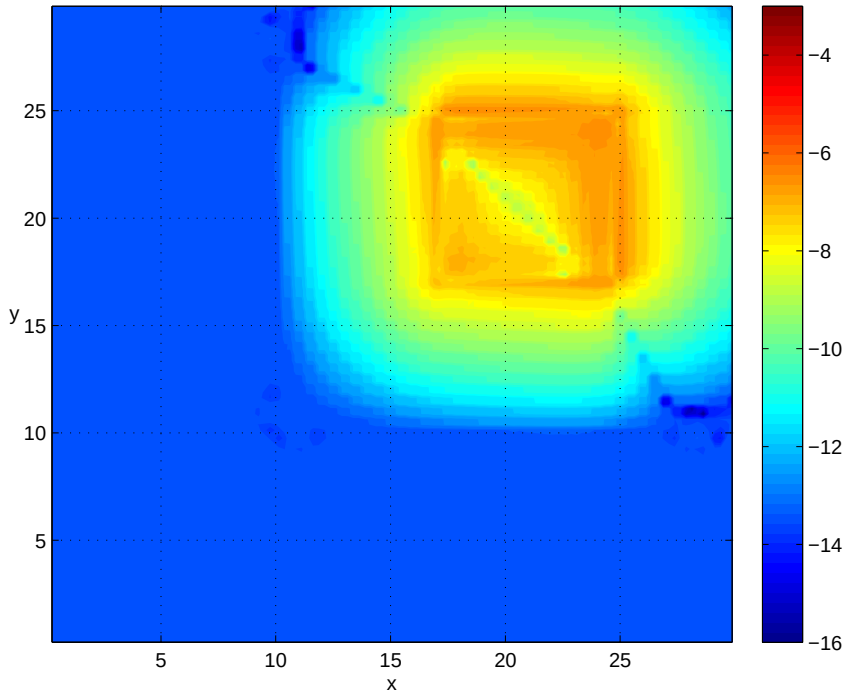
$15.5 \leq x, y \leq 26$



$\max \left|1 - \tilde{\phi}_h/\phi_h\right| = 2.2 \times 10^{-6}$

$max(M^E_{\alpha\beta}, M^{\partial E}_{\alpha\beta}) = 6.2 \times 10^{-6}$

$17 \leq x, y \leq 25$



$\max \left|1 - \tilde{\phi}_h/\phi_h\right| = 3.9 \times 10^{-7}$

$max(M^E_{\alpha\beta}, M^{\partial E}_{\alpha\beta}) = 1.2 \times 10^{-6}$

- Log scale

Summary

- The numerical results showed that this method accurately approximates the transport solution computed on a given grid.
- The metrics of transport effects based the QD tensor can be used to determine diffusion subdomains.
- Domain decomposition methodologies can be formulated using different low-order equations (QD, NDA, DSA)
- The presented methodology can be extended to
  - different transport discretization methods,
  - other multidimensional geometries,
  - multigroup transport problems,
  - time-dependent problems.
- References


N. Stehle, D. Anistratov & M. Adams, “A Hybrid Transport-Diffusion Method for 2D Transport Problems with Diffusive Subdomains," *J. of Computational Physics*, **270**, 325-344, (2014); `http://dx.doi.org/10.1016/j.jcp.2014.03.056`

On Methods of Long Characteristics

Dmitriy Y. Anistratov

Department of Nuclear Engineering
North Carolina State University

NE 723
Neutron Transport Theory
Fall 2026

Outline

1. Method of long characteristics
   - Formulation of the method of long characteristics
   - Conservative method of long characteristics
   - Calculation of the multiplication factor (eigenvalue)
   - Map of characteristics

Transport problem

We consider the one-group transport equation in 2D Cartesian geometry with isotropic scattering and source

$$\vec{\Omega} \cdot \vec{\nabla}\psi(\vec{r},\vec{\Omega}) + \Sigma_t(\vec{r})\psi(\vec{r},\vec{\Omega}) = \frac{1}{4\pi}\Sigma_s(\vec{r}) \int_{4\pi} \psi(\vec{r},\vec{\Omega}')d\vec{\Omega}' + \frac{1}{4\pi}q(\vec{r})\,, \tag{1}$$

$$\vec{\Omega}_m \cdot \vec{\nabla}\psi_m(\vec{r}) + \Sigma_t(\vec{r})\psi_m(\vec{r}) = \frac{1}{4\pi}\Sigma_s(\vec{r}) \sum_{m'} \psi_{m'}(\vec{r})w_{m'} + \frac{1}{4\pi}q(\vec{r})\,, \tag{2}$$

where

$$\psi_m(\vec{r}) = \psi(\vec{r},\vec{\Omega}_m)\,. \tag{3}$$

The transport equation along characteristics

$$\boxed{\frac{d\psi(s)}{ds} + \Sigma_t(s)\psi(s) = Q(s)\,,} \tag{4}$$

$$Q(s) = \frac{1}{4\pi}(\Sigma_s\phi + q)\,, \tag{5}$$

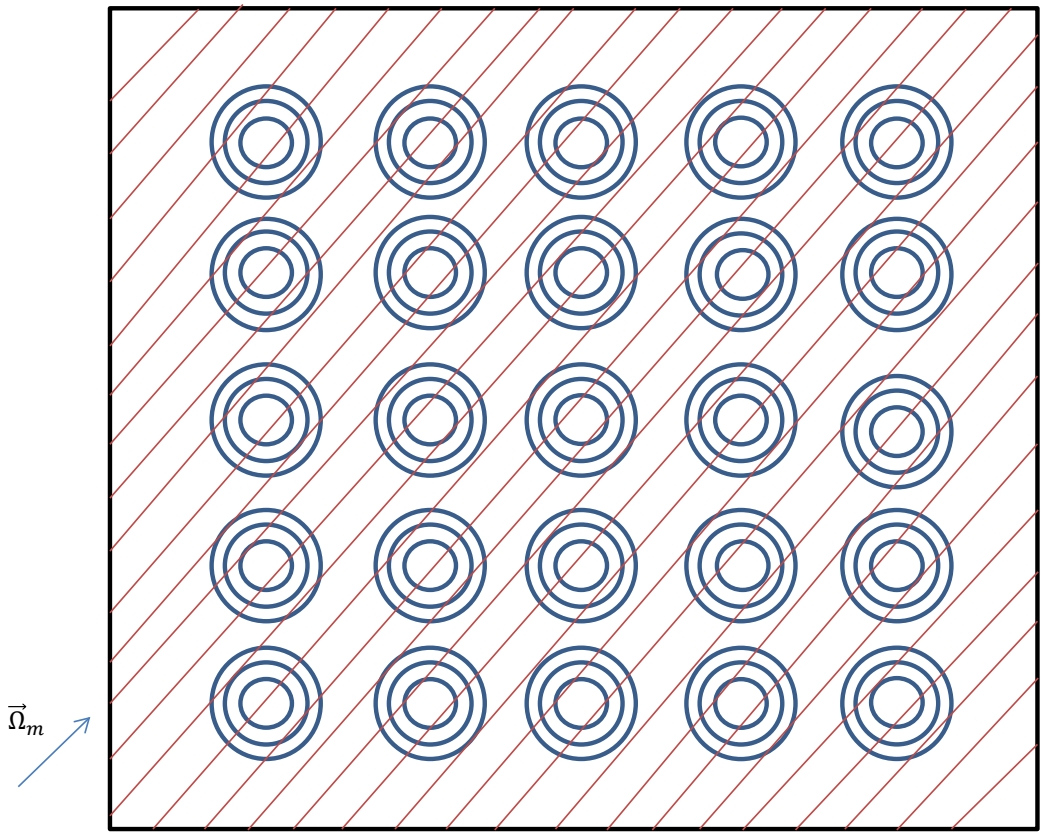
where

$$dx = \Omega_x ds\,, \quad dy = \Omega_y ds\,, \tag{6}$$

$$\Omega_x = \sin\theta\cos\gamma\,, \quad \Omega_y = \sin\theta\sin\gamma\,. \tag{7}$$

Mesh of Characteristics for $\vec{\Omega}_m$

The mesh of characteristics consists of characteristics that cross the whole spatial domain from one external boundary to another external boundary.

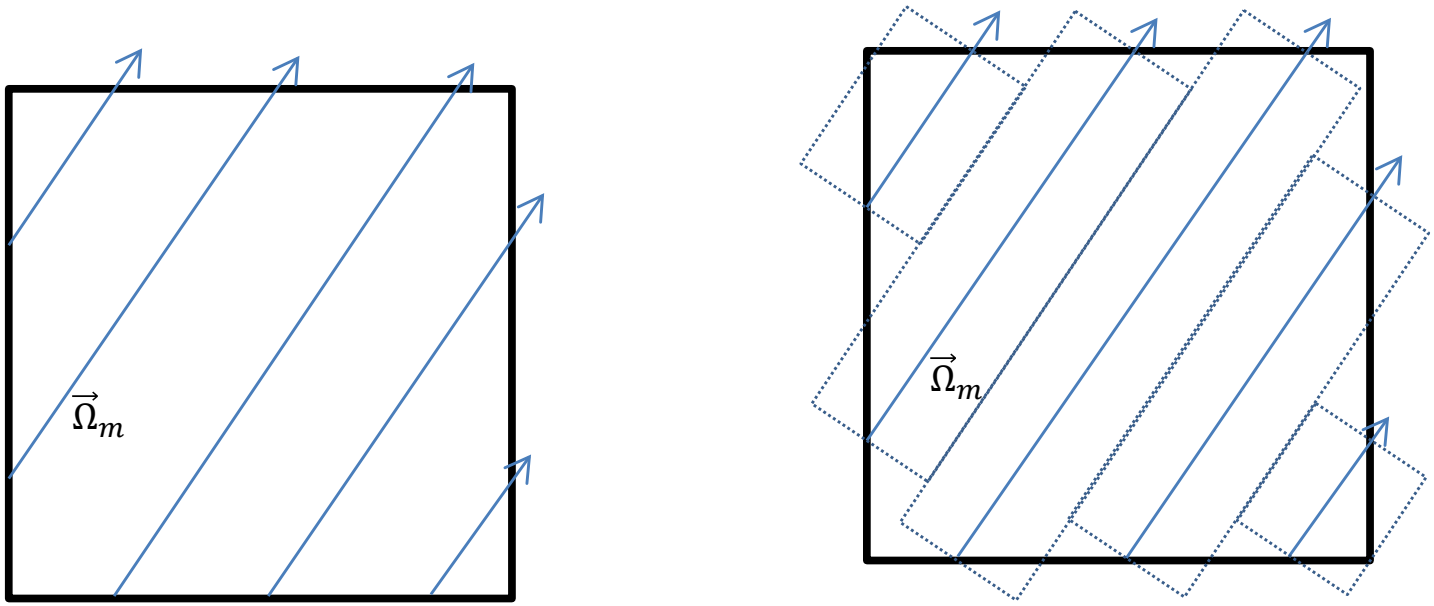


Mesh of **long** characteristics in the direction of $\vec{\Omega}_m$

Computer graphics: **ray-tracing** algorithm

Mesh of characteristics

We now define tubes associated with each characteristics to form a mesh of characteristic tubes.



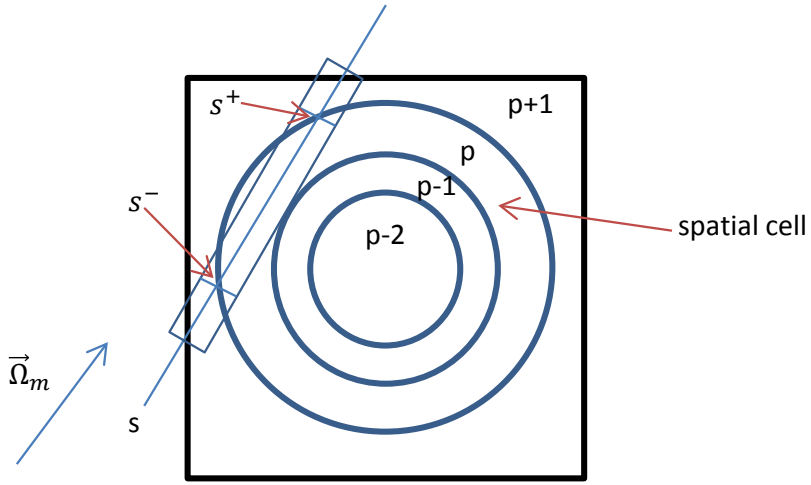
Characteristic mesh and mesh of tubes in a spatial domain.

Equation along characteristics

Integral form of the transport equation along a characteristic

$$\psi(s^+) = \psi(s^-)\exp\left(-\int_{s^-}^{s^+}\Sigma_t(s')ds'\right) + \int_{s^-}^{s^+} Q(s')\exp\left(-\int_{s'}^{s^+}\Sigma_t(s'')ds''\right)ds' . \quad (8)$$

A **spatial cell** is a region with constant cross sections and total source.

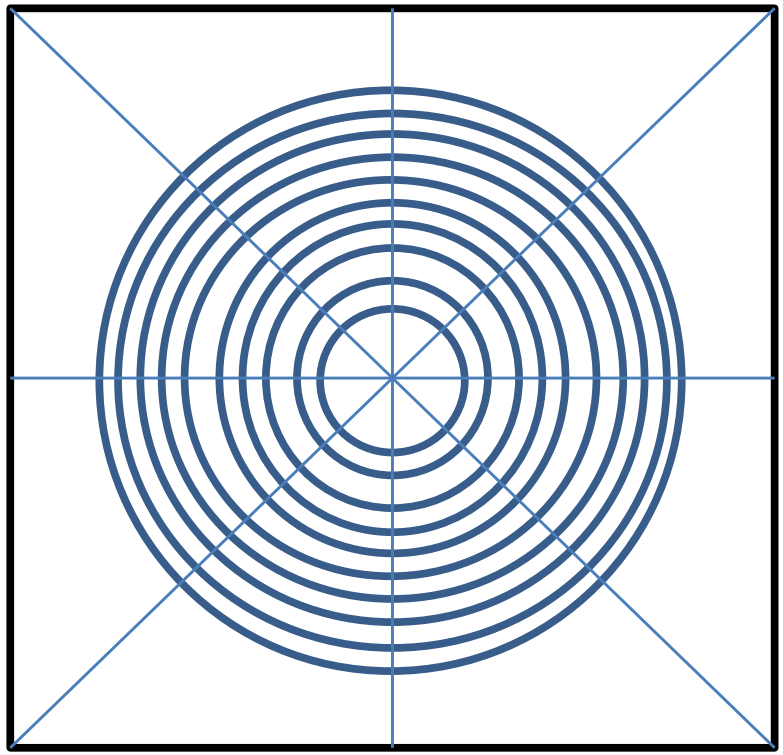


Particle track in a spatial cell.

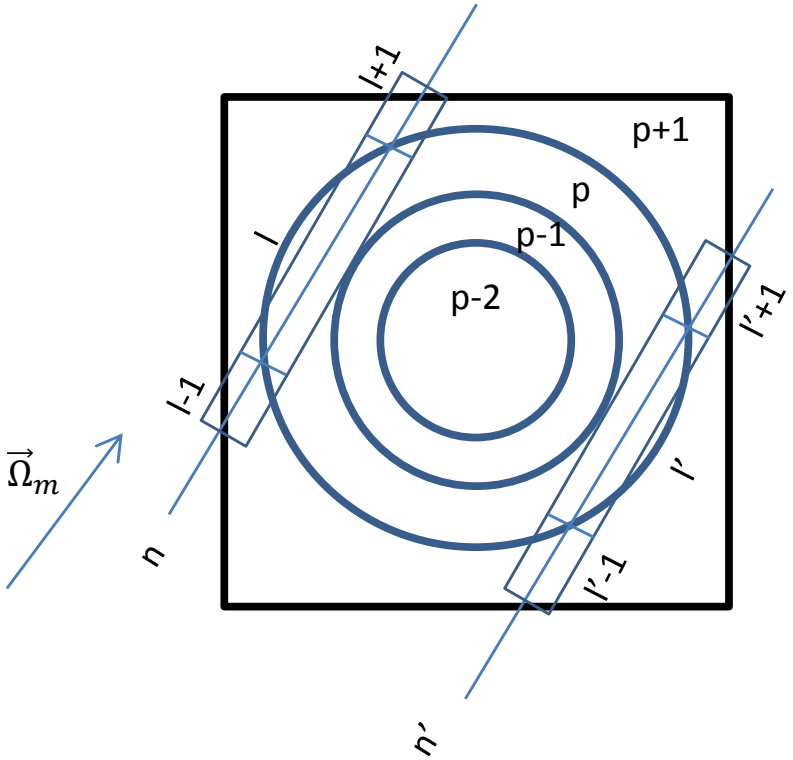
Assuming that cross sections and total source term are constant we get

$$\psi(s^+) = \psi(s^-)e^{-\Sigma_t(s^+-s^-)} + \frac{Q}{\Sigma_t}\left(1 - e^{-\Sigma_t(s^+-s^-)}\right). \quad (9)$$

Material regions in a pin cell

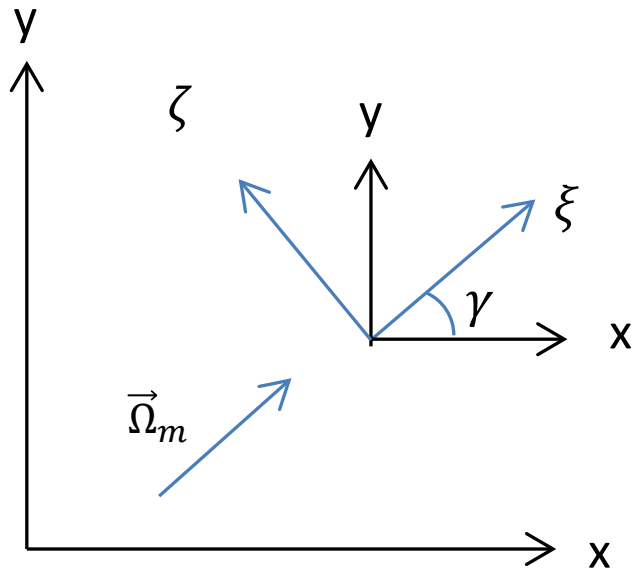


Indexing of spatial cells



- m is the index of the direction of particle flight.
- n is the index of the characteristic in the mesh for $\vec{\Omega}_m$ direction.
- l is the index of the particle track (segment) of the characteristic.
- p is the index of a spatial cell with constant material properties and constant total source.

Local coordinates



We transform (x, y) coordinates to (ξ, ζ), where

$$\vec{e}_\xi = \vec{\Omega}_m\,, \quad \vec{e}_\zeta \cdot \vec{e}_\xi = 0\,. \tag{10}$$

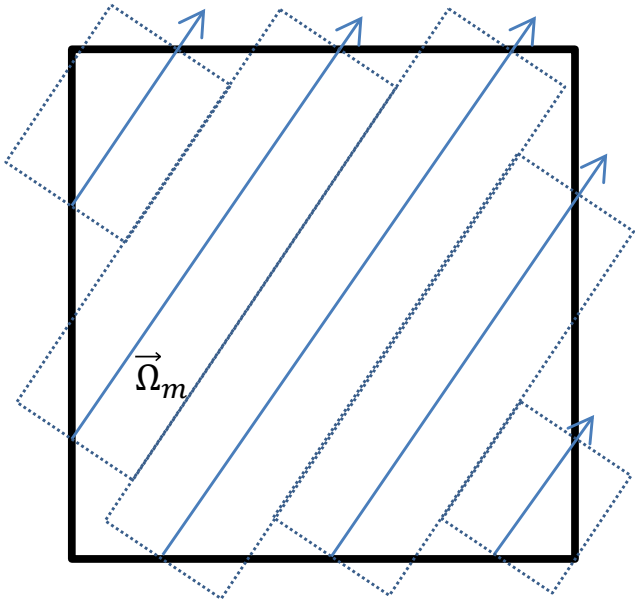
As a result we obtain

$$\vec{\Omega}_m \cdot \vec{\nabla}\psi_m + \Sigma_t \psi_m = Q\,, \tag{11}$$

$$\boxed{\sin\theta_m \frac{\partial \psi_m(\xi,\zeta)}{\partial \xi} + \Sigma_t(\xi,\zeta)\psi_m(\xi,\zeta) = Q(\xi,\zeta)\,.} \tag{12}$$

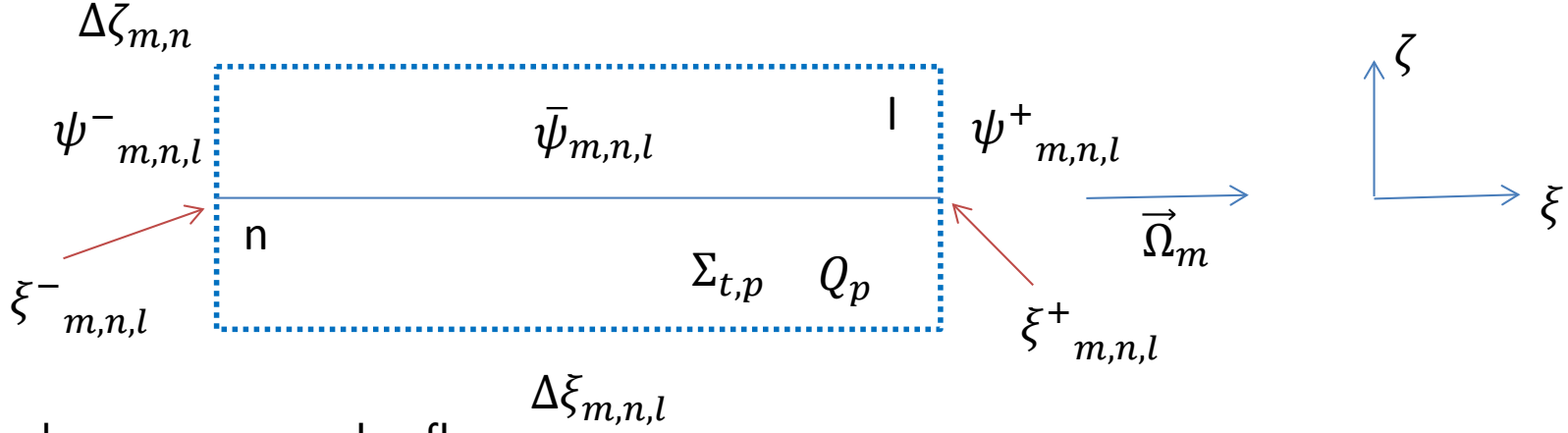
Preservation of the domain area

In general case the area covered by tubes is not equal to the area of the domain.



Characteristic mesh and mesh of associated tubes in the spatial domain.

Segment of the tube associated with the track



The track-average angular flux

$$\bar{\psi}_{m,n,l} = \frac{1}{\Delta\xi_{m,n,l}} \int_{\xi^-_{m,n,l}}^{\xi^+_{m,n,l}} \psi_m(\xi, \zeta_{m,n}) d\xi \,, \tag{13}$$

$\Delta\xi_{m,n,l} = \xi^+_{m,n,l} - \xi^-_{m,n,l}$ is the length of the track.
$\Delta\zeta_{m,n}$ is the width of a tube.
$\Delta\xi^*_{m,n,l} = f_p \Delta\xi_{m,n,l}$ is the modified the track length, where

$$\boxed{f_p = \frac{A_p}{\frac{1}{4\pi} \sum_m w_m \sum_{n,l \in cell\ p} \Delta\xi_{m,n,l} \Delta\zeta_{m,n}}} \,, \tag{14}$$

A_p is the area of p-th cell.

Equations for the tube segment

The equation for the face-average angular flux is given by

$$\psi^{+}_{m,n,l} = \psi^{-}_{m,n,l} \exp\left(-\frac{\Sigma_{t,p}\Delta\xi^{*}_{m,n,l}}{\sin\theta_m}\right) + \frac{Q_p}{\Sigma_{t,p}}\left(1 - \exp\left(-\frac{\Sigma_{t,p}\Delta\xi^{*}_{m,n,l}}{\sin\theta_m}\right)\right). \tag{15}$$

The equation for the track-average angular flux is defined as follows:

$$\bar{\psi}_{m,n,l} = \alpha_{m,n,l}\psi^{-}_{m,n,l} + (1 - \alpha_{m,n,l})\psi^{+}_{m,n,l}\,, \tag{16}$$

$$\alpha_{m,n,l} = \frac{1}{\tau_{m,n,l}} - \frac{e^{-\tau_{m,n,l}}}{1 - e^{-\tau_{m,n,l}}}\,, \tag{17}$$

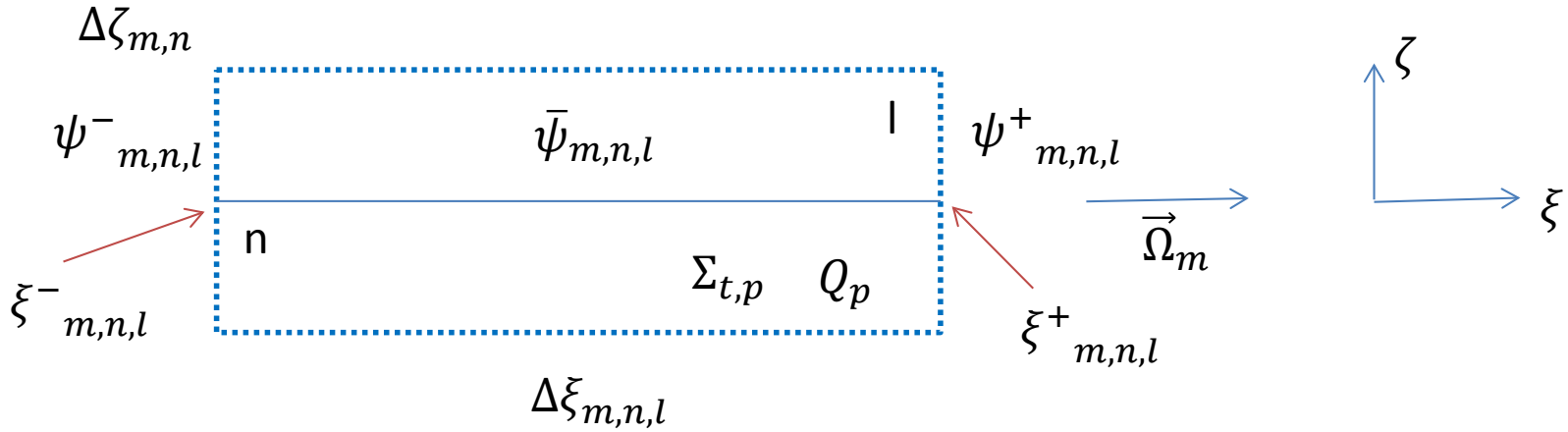
$$\tau_{m,n,l} = \frac{\Sigma_{t,p}\Delta\xi^{*}_{m,n,l}}{\sin\theta_m}\,. \tag{18}$$

The total source term

$$\boxed{Q_p = \frac{1}{4\pi}\left(\Sigma_{s,p}\bar{\phi}_p + q_p\right),} \tag{19}$$

$$\boxed{\bar{\phi}_p = ?\,.} \tag{20}$$

Formulation of particle balance



Detailed particle balance in the segment of the characteristic tube

$$\int_{\Delta\zeta_{m,n}} d\zeta \int_{\xi^-_{m,n,l}(\zeta)}^{\xi^+_{m,n,l}(\zeta)} \left[\sin\theta_m \frac{\partial\psi_m(\xi,\zeta)}{\partial\xi} + \Sigma_t(\xi,\zeta)\psi_m(\xi,\zeta)\right] = \int_{\Delta\zeta_{m,n}} d\zeta \int_{\xi^-_{m,n,l}(\zeta)}^{\xi^+_{m,n,l}(\zeta)} Q(\xi,\zeta)\,.$$

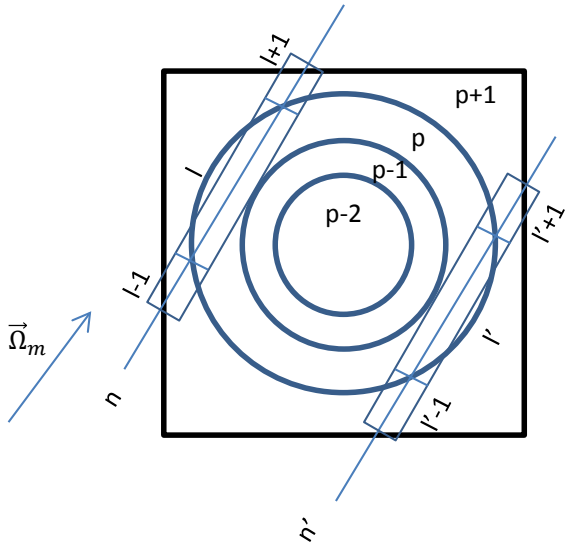
Discretized detailed particle balance in the segment of the characteristic tube

$$\Delta\zeta_{m,n}\left[\sin\theta_m(\psi^+_{m,n,l} - \psi^-_{m,n,l}) + \Sigma_{t,p}\bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\right] = Q_p\xi^*_{m,n,l}\Delta\zeta_{m,n}\,. \qquad (21)$$

Formulation of particle balance

Discretized detailed particle balance in the segment of the characteristic tube

$$\Delta\zeta_{m,n}\left[\sin\theta_m(\psi^+_{m,n,l}-\psi^-_{m,n,l})+\Sigma_{t,p}\bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\right]=Q_p\xi^*_{m,n,l}\Delta\zeta_{m,n}\,. \tag{22}$$



Discretized detailed particle balance in the cell p

$$\sin\theta_m\sum_{n,l\in cell\ p}\Delta\zeta_{m,n}(\psi^+_{m,n,l}-\psi^-_{m,n,l})+\Sigma_{t,p}\sum_{n,l\in cell\ p}\bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\Delta\zeta_{m,n}=$$

$$Q_p\sum_{n,l\in cell\ p}\xi^*_{m,n,l}\Delta\zeta_{m,n}\,. \tag{23}$$

Leakage rate density in the cell p per steradian

Formulation of particle balance

Discretized detailed particle balance in the cell p

$$\sin\theta_m \sum_{n,l\in cell\ \ p} \Delta\zeta_{m,n}(\psi^+_{m,n,l} - \psi^-_{m,n,l}) + \Sigma_{t,p} \sum_{n,l\in cell\ \ p} \bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\Delta\zeta_{m,n} = Q_p \sum_{n,l\in cell\ \ p} \xi^*_{m,n,l}\Delta\zeta_{m,n}\,. \quad (26)$$

Leakage rate density in the cell p per steradian

$$\boxed{\sin\theta_m \sum_{n,l\in cell\ \ p} \Delta\zeta_{m,n}(\psi^+_{m,n,l} - \psi^-_{m,n,l})\,.} \quad (27)$$

Collision rate density in the cell p per steradian

$$\boxed{\Sigma_{t,p} \sum_{n,l\in cell\ \ p} \bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\Delta\zeta_{m,n}\,.} \quad (28)$$

Scattering rate density and *group source rate density* in the cell p per steradian

$$\boxed{Q_p \sum_{n,l\in cell\ \ p} \xi^*_{m,n,l}\Delta\zeta_{m,n}\,,} \quad \text{where} \quad Q_p = \frac{1}{4\pi}\left(\Sigma_{s,p}\bar{\phi}_p + q_p\right). \quad (29)$$

Formulation of particle balance

Particle balance in the cell p

$$\sum_m w_m \sin\theta_m \sum_{n,l\in cell\ p} \Delta\zeta_{m,n}(\psi^+_{m,n,l}-\psi^-_{m,n,l})+\Sigma_{t,p}\sum_m w_m \sum_{n,l\in cell\ p} \bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\Delta\zeta_{m,n} = Q_p \sum_m w_m \sum_{n,l\in cell\ p} \xi^*_{m,n,l}\Delta\zeta_{m,n}\,. \quad (30)$$

Net leakage rate (LR) in the cell p

$$\boxed{LR_p = \sum_m w_m \sin\theta_m \sum_{n,l\in cell\ p} \Delta\zeta_{m,n}(\psi^+_{m,n,l}-\psi^-_{m,n,l})\,,} \quad (31)$$

$$LR_p = LR^+_p - LR^-_p\,, \quad (32)$$

$$LR^\pm_p = \sum_m w_m \sin\theta_m \sum_{n,l\in cell\ p} \Delta\zeta_{m,n}\psi^\pm_{m,n,l}\,. \quad (33)$$

Collision rate in the cell p

$$\boxed{\Sigma_{t,p}\sum_m w_m \sum_{n,l\in cell\ p} \bar{\psi}_{m,n,l}\Delta\xi^*_{m,n,l}\Delta\zeta_{m,n}\,.} \quad (34)$$

Formulation of particle balance

Particle balance in the cell p

$$LR_p + \Sigma_{t,p} \boxed{\sum_m w_m \sum_{n,l \in cell\ p} \Delta\xi^*_{m,n,l} \Delta\zeta_{m,n} \bar{\psi}_{m,n,l}} =$$

$$\Sigma_{s,p} \boxed{\frac{1}{4\pi} \bar{\phi}_p \sum_m w_m \sum_{n,l \in cell\ p} \xi^*_{m,n,l} \Delta\zeta_{m,n}} + \frac{1}{4\pi} q_p \sum_m w_m \sum_{n,l \in cell\ p} \xi^*_{m,n,l} \Delta\zeta_{m,n} \,. \quad (35)$$

The cell-average scalar flux

$$\bar{\phi}_p = \frac{\sum_m w_m \sum_{n,l \in cell\ p} \Delta\xi^*_{m,n,l} \Delta\zeta_{m,n} \bar{\psi}_{m,n,l}}{\frac{1}{4\pi} \sum_m w_m \sum_{n,l \in cell\ p} \xi^*_{m,n,l} \Delta\zeta_{m,n}} = \boxed{\frac{1}{A_p} \sum_m w_m \sum_{n,l \in cell\ p} \Delta\xi^*_{m,n,l} \Delta\zeta_{m,n} \bar{\psi}_{m,n,l}} \,. \quad (36)$$

Note that

$$\frac{1}{4\pi} \sum_m w_m \sum_{n,l \in cell\ p} \xi^*_{m,n,l} \Delta\zeta_{m,n} = A_p \,, \quad (37)$$

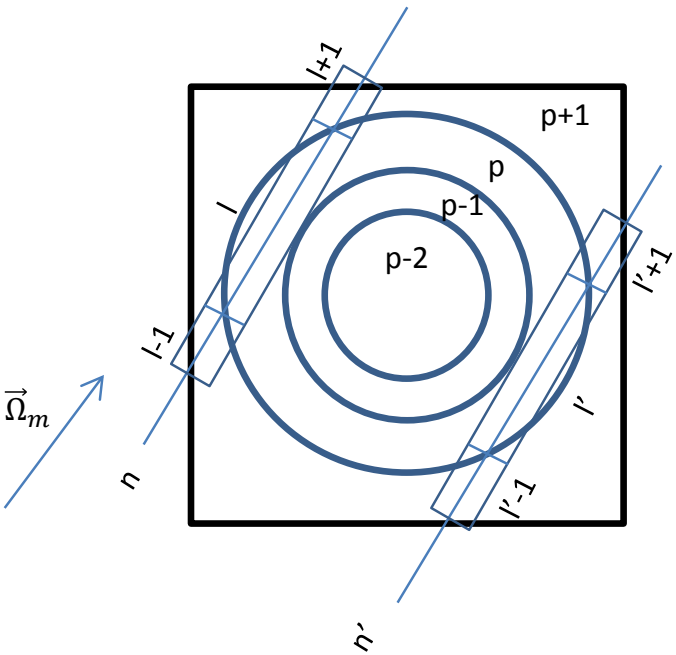
Particle balance equation in the cell p

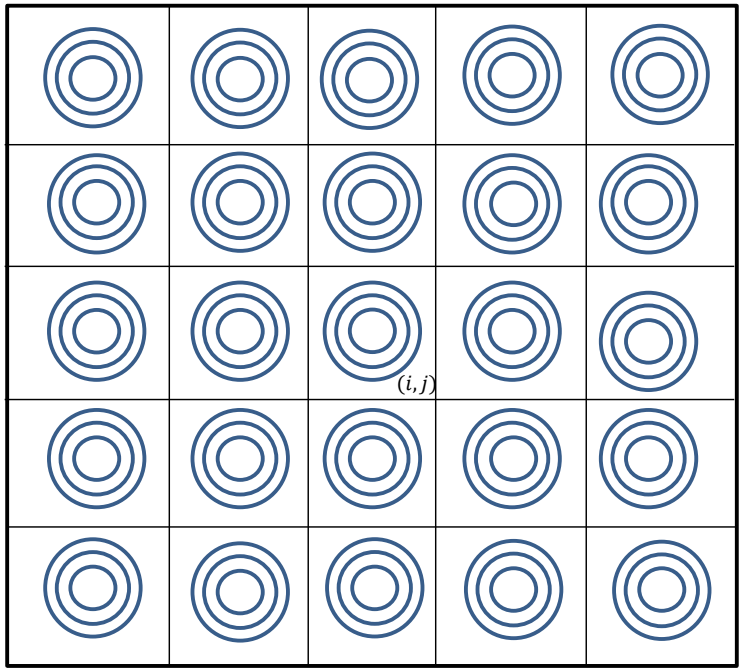
$$LR_p + \Sigma_{t,p} A_p \bar{\phi}_p = \left(\Sigma_{s,p} \bar{\phi}_p + q_p \right) A_p \,. \quad (38)$$

Particle balance in the problem domain

Particle balance in cell p

$$LR_p + \Sigma_{a,p}\bar{\phi}_p A_p = \frac{1}{k}\nu_{f,p}\Sigma_{f,p}\bar{\phi}_p A_p \,. \tag{39}$$





Particle balance in pin cell (i,j)

$$\sum_{p\in pin\ \ cell\ \ i,j} LR_p + \sum_{p\in pin\ \ cell\ \ i,j} \Sigma_{a,p}\bar{\phi}_p A_p = \frac{1}{k}\sum_{p\in pin\ \ cell\ \ i,j} \nu_{f,p}\Sigma_{f,p}\bar{\phi}_p A_p \,. \tag{40}$$

Particle balance in the problem domain

Particle balance in pin cell (i,j)

$$\sum_{p \in pin\ \ cell\ \ i,j} LR_p + \sum_{p \in pin\ \ cell\ \ i,j} \Sigma_{a,p} \bar{\phi}_p A_p = \frac{1}{k} \sum_{p \in pin\ \ cell\ \ i,j} \nu_{f,p} \Sigma_{f,p} \bar{\phi}_p A_p \,. \tag{41}$$

Net leakage rate in the pin cell (i,j)

$$LR_{i,j}^{pin\ cell} = \sum_{p \in pin\ \ cell\ \ i,j} LR_p \,. \tag{42}$$

Absorption rate in the pin cell (i,j)

$$AR_{i,j}^{pin\ cell} = \sum_{p \in pin\ \ cell\ \ i,j} \Sigma_{a,p} \bar{\phi}_p A_p \,. \tag{43}$$

Production rate in the pin cell (i,j)

$$PR_{i,j}^{pin\ cell} = \sum_{p \in pin\ \ cell\ \ i,j} \nu_{f,p} \Sigma_{f,p} \bar{\phi}_p A_p \,. \tag{44}$$

Particle balance in the pin cell (i,j)

$$LR_{i,j}^{pin\ cell} + AR_{i,j}^{pin\ cell} = \frac{1}{k} PR_{i,j}^{pin\ cell} \,. \tag{45}$$

Particle Balance in the Problem Domain

Particle balance in the pin cell (i, j)

$$LR_{i,j}^{\text{pin cell}} + AR_{i,j}^{\text{pin cell}} = \frac{1}{k} PR_{i,j}^{\text{pin cell}} . \tag{46}$$

Particle balance in the problem domain

$$\sum_{i,j} LR_{i,j}^{\text{pin cell}} + \sum_{i,j} AR_{i,j}^{\text{pin cell}} = \frac{1}{k} \sum_{i,j} PR_{i,j}^{\text{pin cell}} , \tag{47}$$

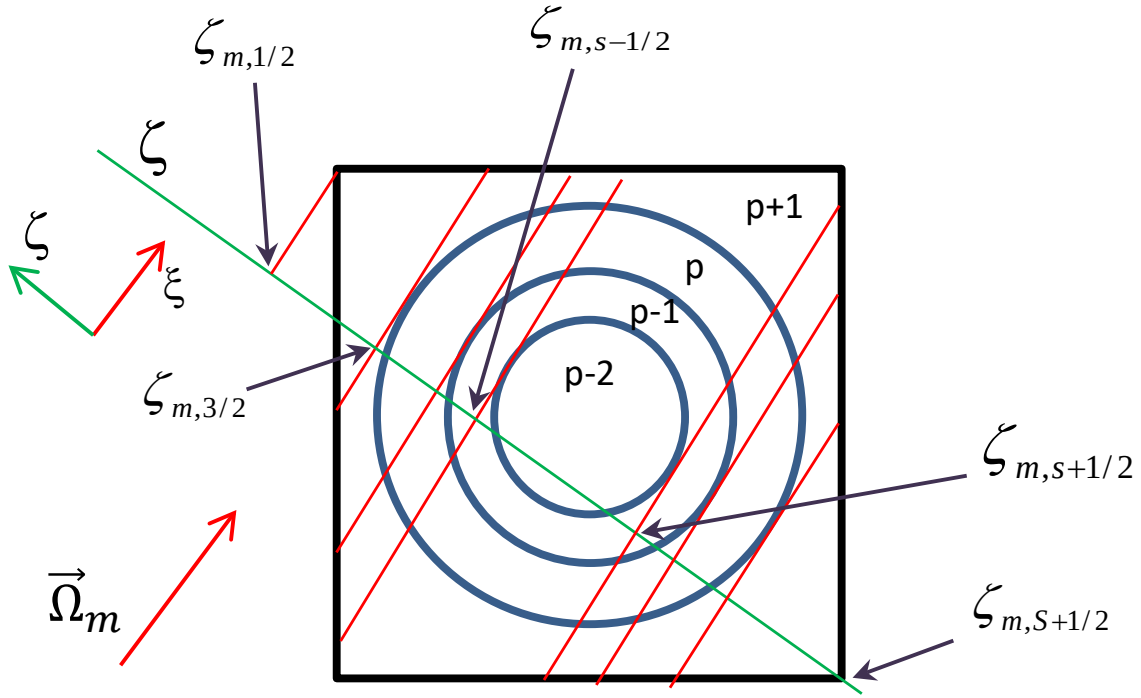
$$LR + AR = \frac{1}{k} PR , \tag{48}$$

$$LR = \sum_{i,j} LR_{i,j}^{\text{pin cell}} , \quad AR = \sum_{i,j} AR_{i,j}^{\text{pin cell}} , \quad PR = \sum_{i,j} PR_{i,j}^{\text{pin cell}} . \tag{49}$$

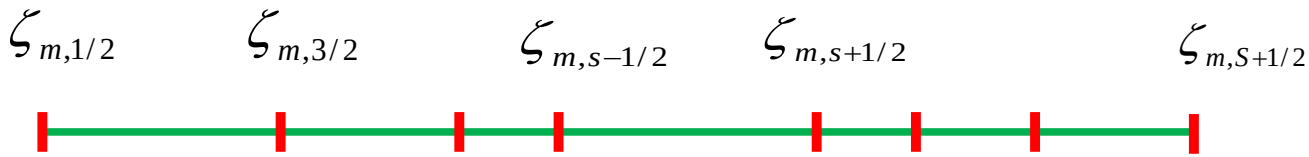
The multiplication factor

$$k = \frac{PR}{LR + AR} . \tag{50}$$

Mapping

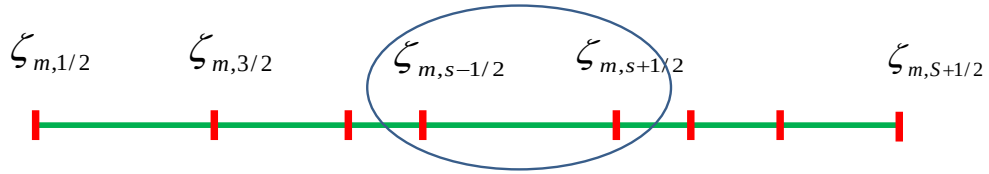


Material subdomains of the characteristic mesh



Map of spatial intervals for $\vec{\Omega}_m$

Mapping of characteristic mesh for $\vec{\Omega}_m$



- Given d^c is the target distance between characteristics
- S is the number of subdomains

$$\Delta\zeta_{m,s} = \zeta_{m,s+1/2} - \zeta_{m,s-1/2}\,, \quad s = 1, \ldots, S$$

- Number of characteristics that can fit in the subdomain s for the direction m

$$\tilde{N}^c_{m,s} = \left[\frac{\Delta\zeta_{m,s}}{d^c}\right] + 1$$

Notations: $[\alpha]$ is the integer part of α

- Determine the actual distance between characteristics in the subdomain

$$d^c_{m,s} = \frac{\Delta\zeta_{m,s}}{\tilde{N}^c_{m,s}}$$

- The mesh of characteristics in the subdomain s for the direction m

$$\tilde{\zeta}^{m,s}_1 = \zeta_{m,s-1/2} + 0.5 d^c_{m,s}$$

$$\tilde{\zeta}^{m,s}_j = \tilde{\zeta}^{m,s}_{j-1} + d^c_{m,s}\,, \quad j = 2, \ldots, \tilde{N}^c_{m,s}$$

$\tilde{\zeta}^{m,s}_j$ is the position of characteristics in the subdomain

On Methods of Short Characteristics

Dmitriy Y. Anistratov

Department of Nuclear Engineering
North Carolina State University

NE 723
Neutron Transport Theory
Fall 2026

Outline

1. Method of short characteristics
   - Formulation of the method of short characteristics
   - Method of short characteristics with different interpolation schemes
   - Conservative method of short characteristics

Transport problem

We consider the one-group transport equation in 2D Cartesian geometry with isotropic scattering and source

$$\vec{\Omega} \cdot \vec{\nabla}\psi(\vec{r},\vec{\Omega}) + \Sigma_t(\vec{r})\psi(\vec{r},\vec{\Omega}) = \frac{1}{4\pi}\Sigma_s(\vec{r}) \int_{4\pi} \psi(\vec{r},\vec{\Omega}')d\vec{\Omega}' + \frac{1}{4\pi} q(\vec{r}) \,, \tag{1}$$

$$\vec{\Omega}_m \cdot \vec{\nabla}\psi_m(\vec{r}) + \Sigma_t(\vec{r})\psi_m(\vec{r}) = \frac{1}{4\pi}\Sigma_s(\vec{r}) \sum_{m'} \psi_{m'}(\vec{r}) w_{m'} + \frac{1}{4\pi} q(\vec{r}) \,, \tag{2}$$

where

$$\psi_m(\vec{r}) = \psi(\vec{r},\vec{\Omega}_m) \,. \tag{3}$$

The transport equation along characteristics

$$\frac{d\psi(s)}{ds} + \Sigma_t(s)\psi(s) = Q(s) \,, \tag{4}$$

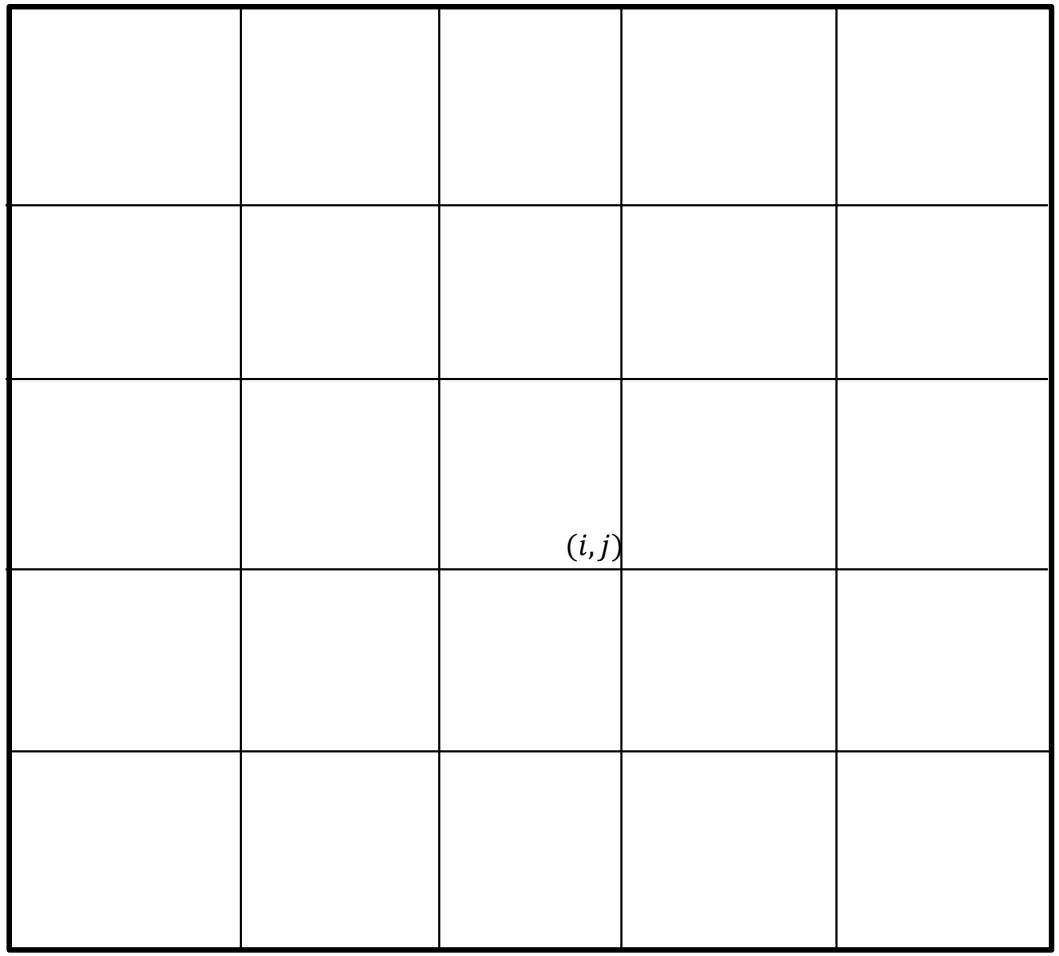
$$Q(s) = \frac{1}{4\pi}(\Sigma_s\phi + q) \,, \tag{5}$$

where

$$dx = \Omega_x ds \,, \quad dy = \Omega_y ds \,, \tag{6}$$

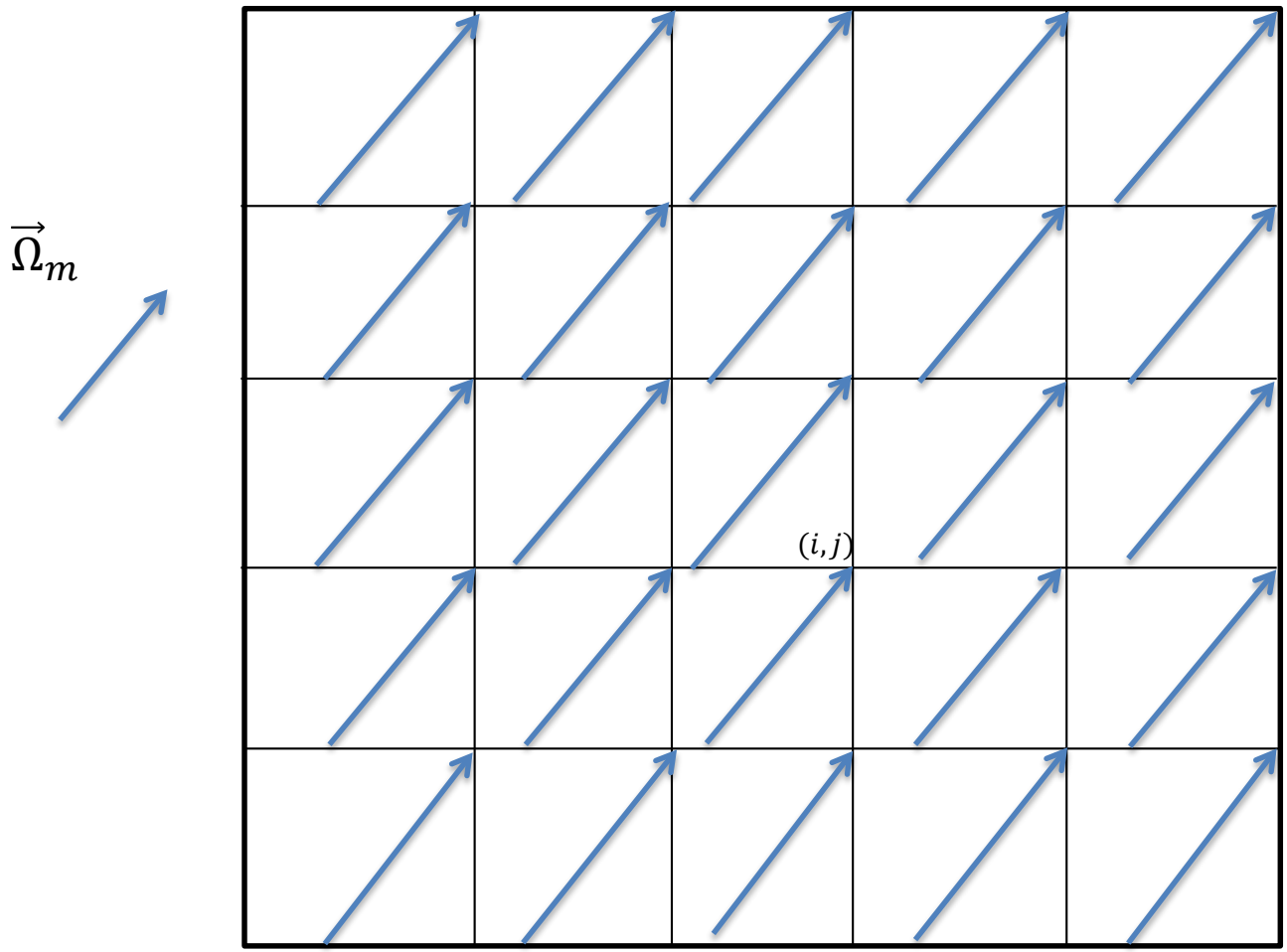
$$\Omega_x = \sin\theta\cos\gamma \,, \quad \Omega_y = \sin\theta\sin\gamma \,. \tag{7}$$

Method of short characteristics (MOSC)

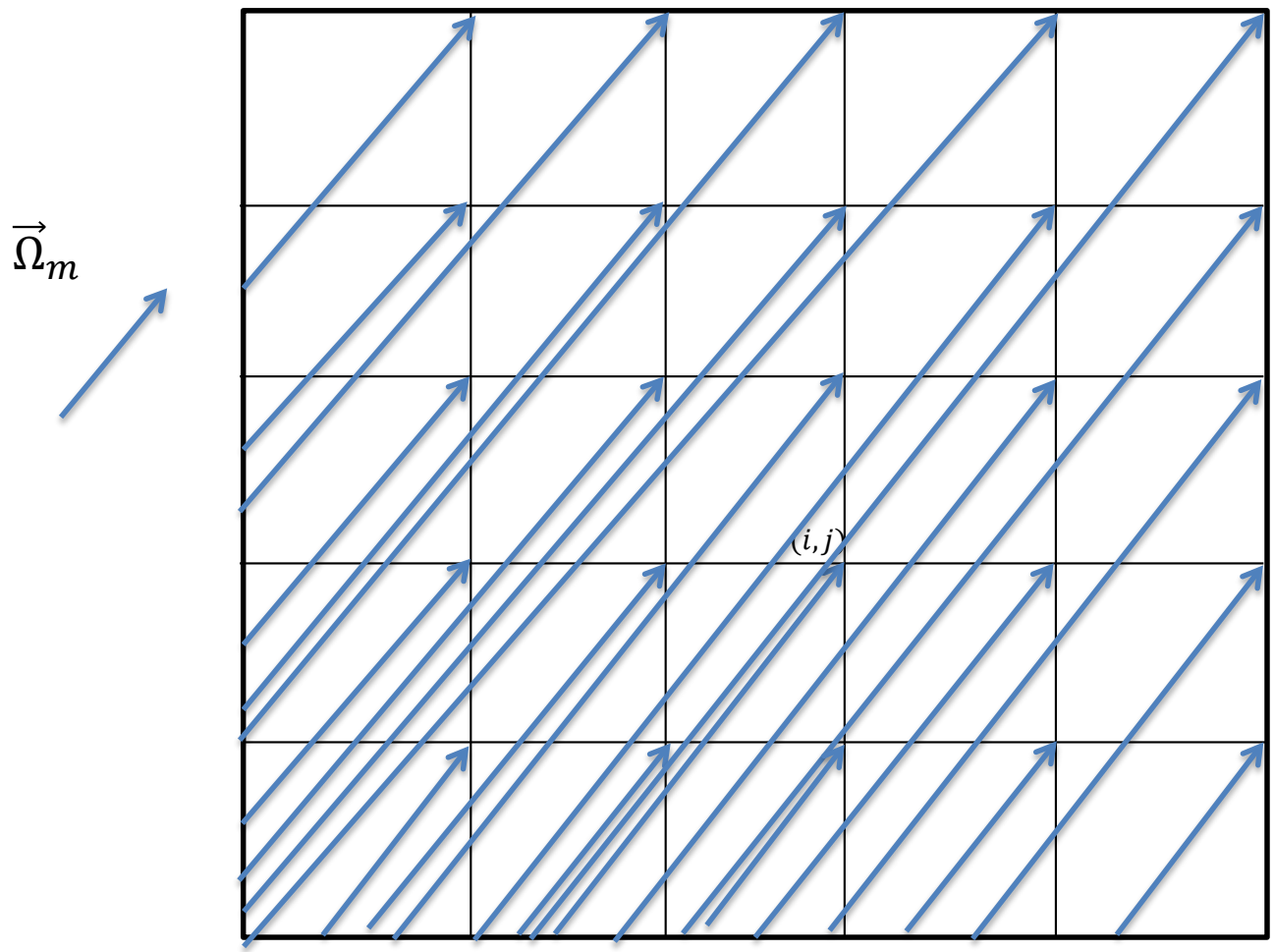


Method of short characteristics (MOSC)

- MOSC $\equiv$ method of characteristics with interpolation



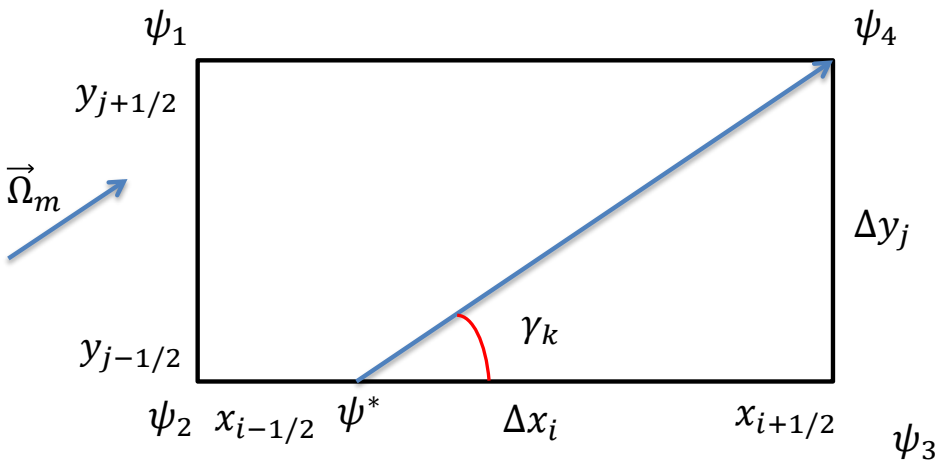
Method of long characteristics



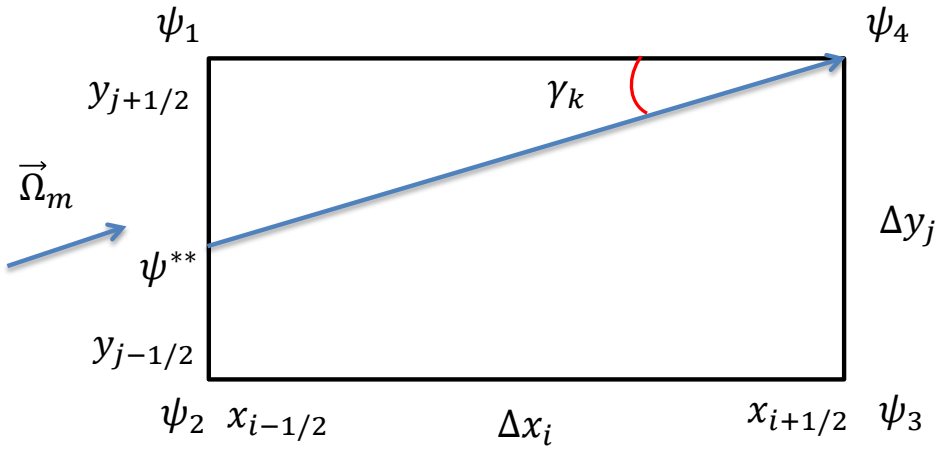
Method of short characteristics (MOSC)

- Let us consider a vertex-based method of short characteristics.
- Known values at vertices: ψ_1, ψ_2, ψ_3
- Unknown value: ψ_4

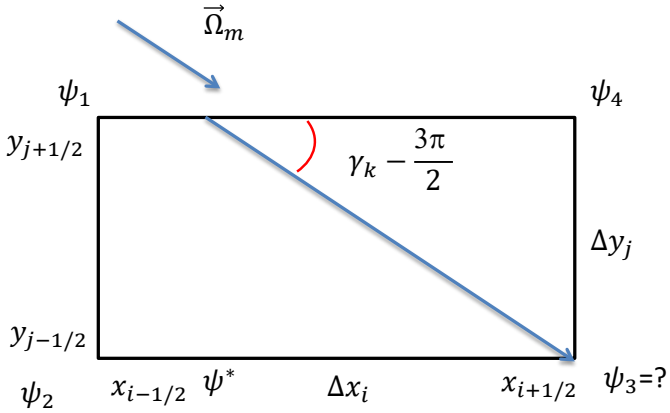
$$\xi_{ijk} = \frac{\Delta y_j}{\Delta x_i} \cot \gamma_k < 1$$

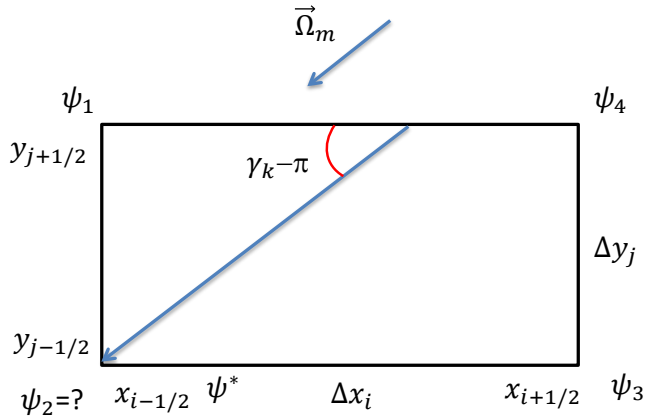


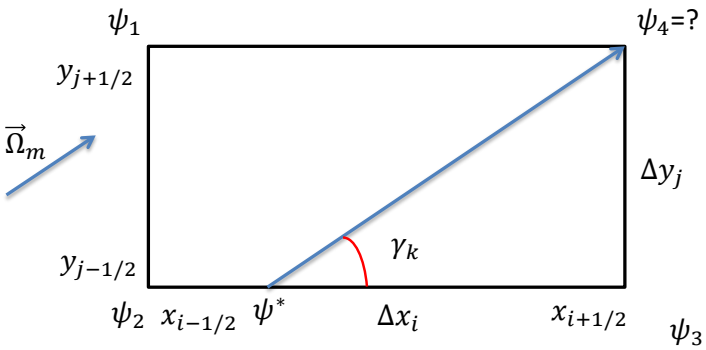
$$\xi_{ijk} = \frac{\Delta y_j}{\Delta x_i} \cot \gamma_k > 1$$

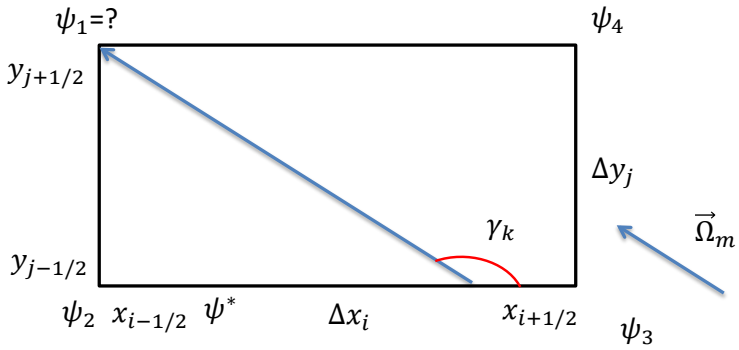


Mesh sweeps


$\vec{\Omega}_m$
ψ_1
ψ_4
$y_{j+1/2}$
$\gamma_k - \frac{3\pi}{2}$
Δy_j
$y_{j-1/2}$
ψ_2
$x_{i-1/2}$
ψ^*
Δx_i
$x_{i+1/2}$
ψ_3=?


$\vec{\Omega}_m$
ψ_1
ψ_4
$y_{j+1/2}$
$\gamma_k - \pi$
Δy_j
$y_{j-1/2}$
ψ_2=?
$x_{i-1/2}$
ψ^*
Δx_i
$x_{i+1/2}$
ψ_3


ψ_1
ψ_4=?
$y_{j+1/2}$
$\vec{\Omega}_m$
Δy_j
γ_k
$y_{j-1/2}$
ψ_2
$x_{i-1/2}$
ψ^*
Δx_i
$x_{i+1/2}$
ψ_3


ψ_1=?
ψ_4
$y_{j+1/2}$
Δy_j
γ_k
$\vec{\Omega}_m$
$y_{j-1/2}$
ψ_2
$x_{i-1/2}$
ψ^*
Δx_i
$x_{i+1/2}$
ψ_3

Mesh sweeps

$\Omega_x > 0 \quad \Omega_y < 0$

$\vec{\Omega}_m$ 1	2	3	4	5
6	7	8	9	10
11	12	13 (i,j)	14	15
16	17	18	19	20
21	22	23	24	25

$\Omega_x < 0 \quad \Omega_y < 0$

5	4	3	2	$\vec{\Omega}_m$ 1
10	9	8	7	6
15	14	13 (i,j)	12	11
20	19	18	17	16
25	24	23	22	21

$\Omega_x > 0 \quad \Omega_y > 0$

21	22	23	24	25
16	17	18	19	20
11	12	13 (i,j)	14	15
6	7	8	9	10
$\vec{\Omega}_m$ 1	2	3	4	5

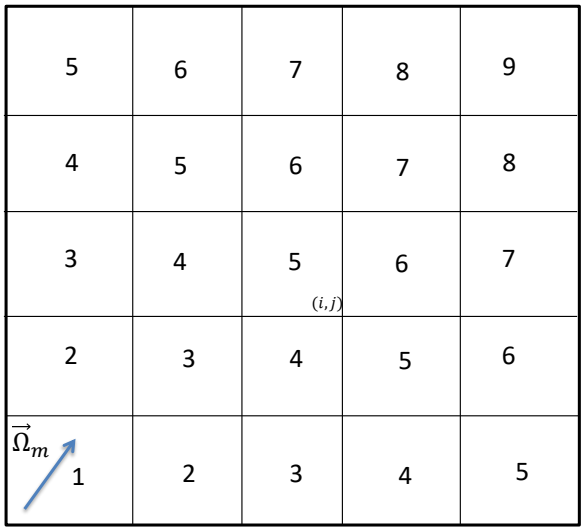
$\Omega_x < 0 \quad \Omega_y > 0$

25	24	23	22	21
20	19	18	17	16
15	14	13 (i,j)	12	11
10	9	8	7	6
5	4	3	2	1 $\vec{\Omega}_m$

Mesh sweeps in parallel

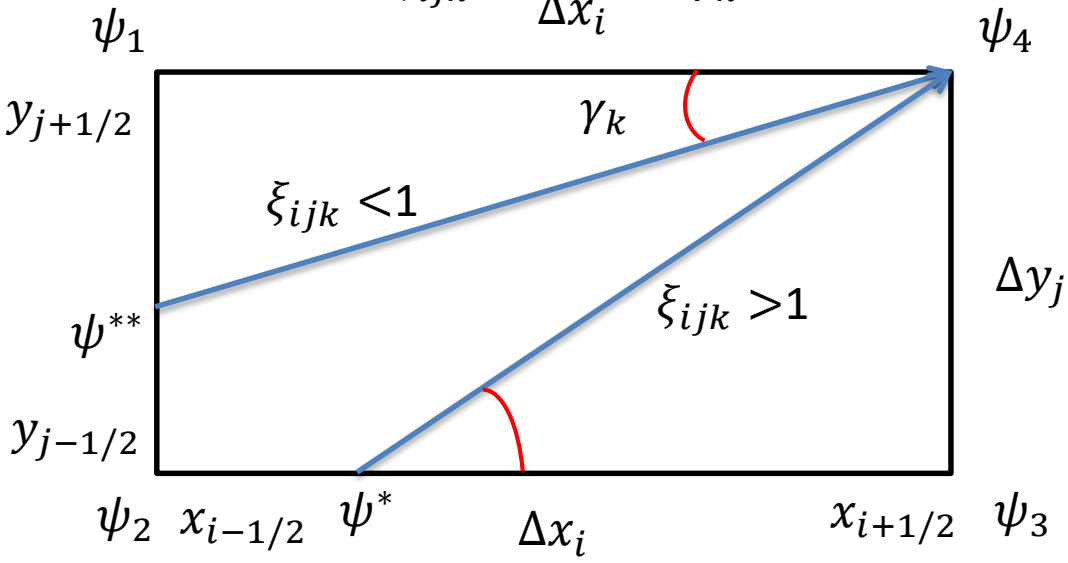
- Koch-Baker-Alcouffe (KBA) algorithm

$$\Omega_x > 0 \quad \Omega_y > 0$$



Method of short characteristics with linear interpolation

$$\xi_{ijk} = \frac{\Delta y_j}{\Delta x_i} \cot \gamma_k$$



$$\tau_{ijkm} = \Sigma_{t,ij} \Delta s_{ijkm} = \frac{\Sigma_{t,ij} \Delta \tilde{s}_{ijk}}{\sin \theta_m}$$

Case $\xi_{ijk} \leq 1$

$$\Delta \tilde{s}_{ijk} = \frac{\Delta y_j}{\sin \gamma_k}$$

$$\psi_4 = \psi^* e^{-\tau_{ijkm}} + \frac{Q_{ij}}{\Sigma_{t,ij}} \left(1 - e^{-\tau_{ijkm}}\right)$$

$$\psi^* = \xi_{ijk} \psi_2 + (1 - \xi_{ijk}) \psi_3$$

Case $\xi_{ijk} > 1$

$$\Delta \tilde{s}_{ijk} = \frac{\Delta x_i}{\cos \gamma_k}$$

$$\psi_4 = \psi^{**} e^{-\tau_{ijkm}} + \frac{Q_{ij}}{\Sigma_{t,ij}} \left(1 - e^{-\tau_{ijkm}}\right).$$

$$\psi^{**} = \frac{1}{\xi_{ijk}} \psi_2 + (1 - \frac{1}{\xi_{ijk}}) \psi_1$$